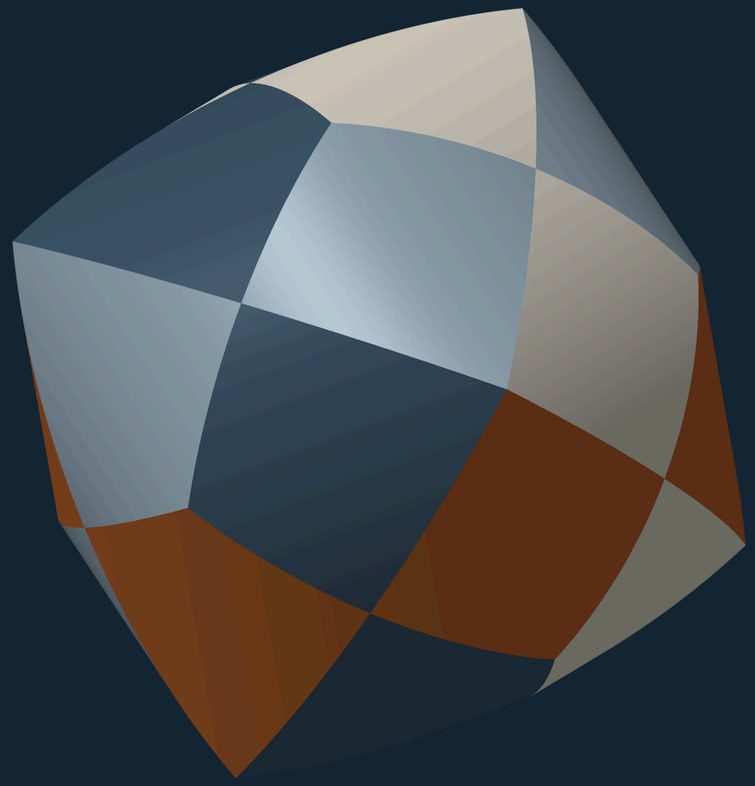
NLD

NONLINEAR DYNAMICS LABORATORY



# Differentiating and integrating measured motion

Model-free kinematics from one sensor or two

github.com/nld-uri/kinematics-toolbox · doi:10.5281/zenodo.23549236

Nonlinear Dynamics Laboratory · University of Rhode Island

THE PROBLEM

# Two problems every vibration lab has

*The setting: one sensor type, and you need the other kinematic quantities*

**You measured displacement (laser vibrometer, DIC, encoder) and you need velocity and acceleration.**

Restoring-force surfaces, NNM energy balance, damping identification, model updating.

**You measured acceleration (the cheap, rugged, ubiquitous sensor) and you need velocity and displacement.**

Mode shapes in physical units, drift-free response, base-excitation compensation, fatigue cycles.

Both are routine requests. Both are harder than they look.

The naive answers — finite differences one way, cumulative trapezoid the other — fail, and fail differently.

THE PROBLEM

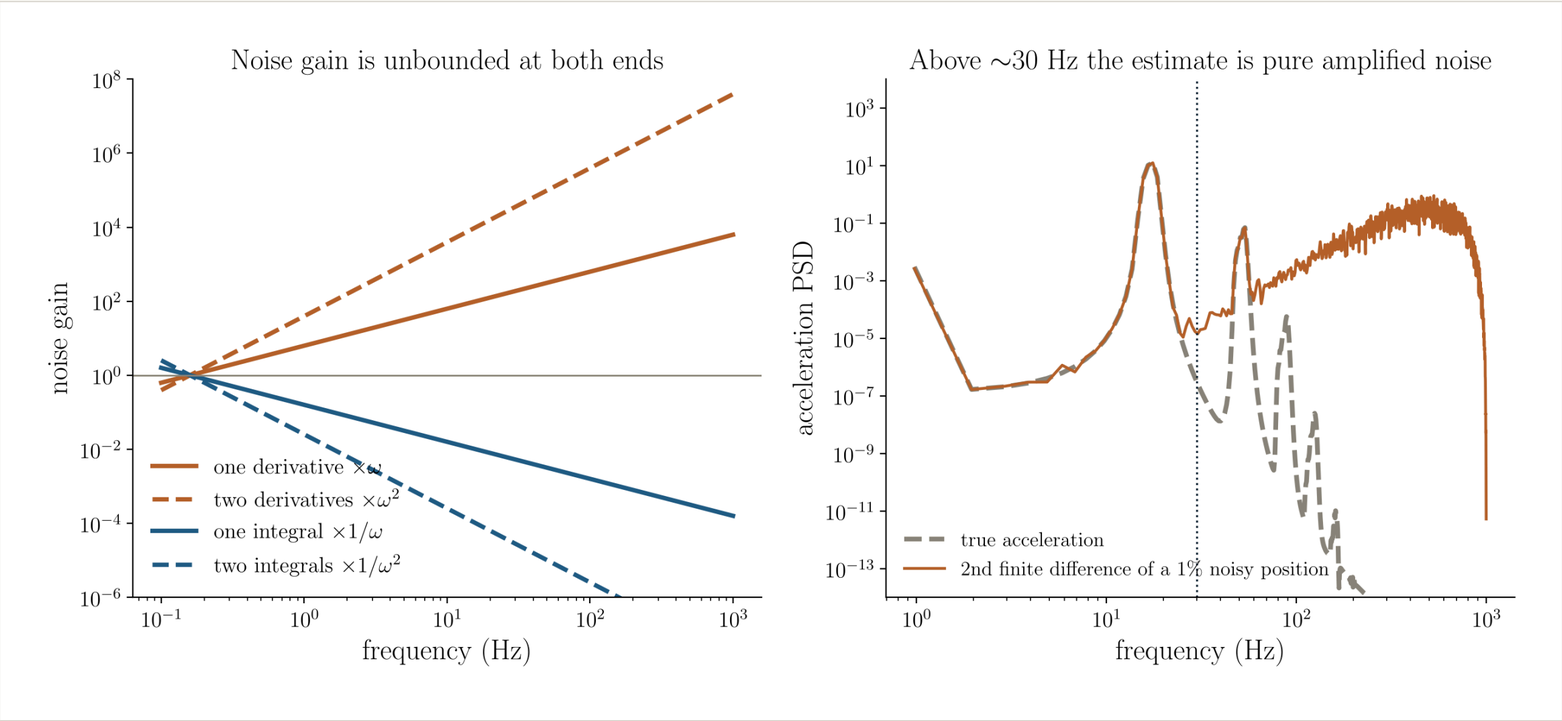
# They are the same problem, run in opposite directions

*One framing that organizes the whole landscape*

- Write the measurement as $y = Au + e$, with $A$ the integration operator.
- Differentiation: $u$ is the derivative you want, $A$ integrates it back to what you measured.
- Integration: $u$ is the displacement, $A$ integrates the acceleration twice.
- **$A$ is compact. Its inverse is unbounded.** This is the definition of an ill-posed inverse problem.
- Noise is amplified without limit — by $\omega$ per derivative going up, by $1/\omega$ per integration going down.
- No amount of cleverness removes this. It is a property of the operator, not of your algorithm.
- Every usable method therefore adds a regularity prior. That is the only thing that distinguishes them.

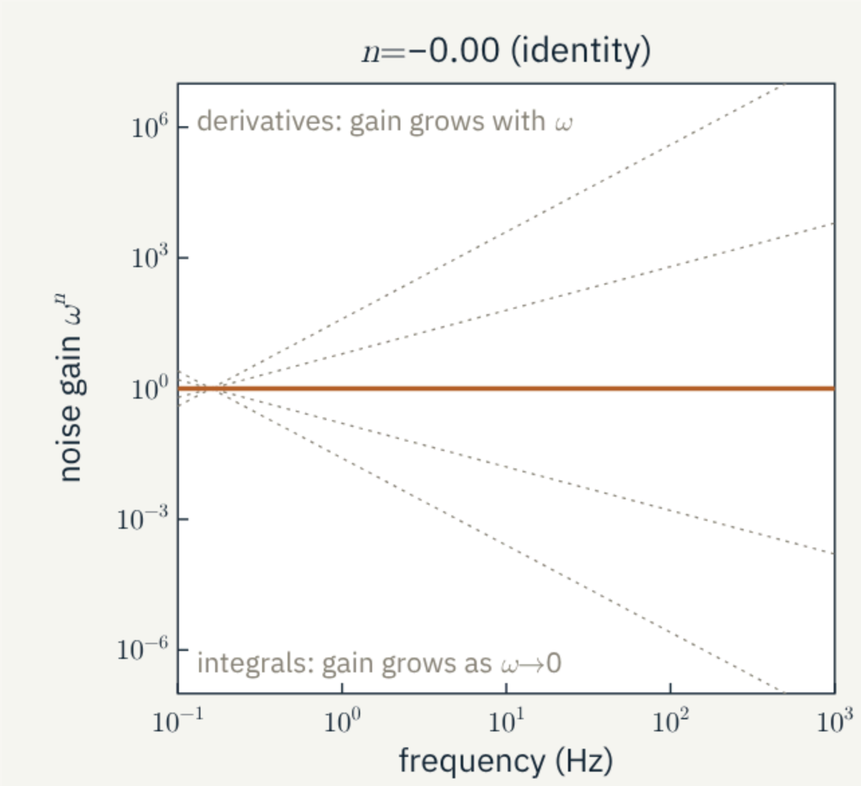
THE PROBLEM

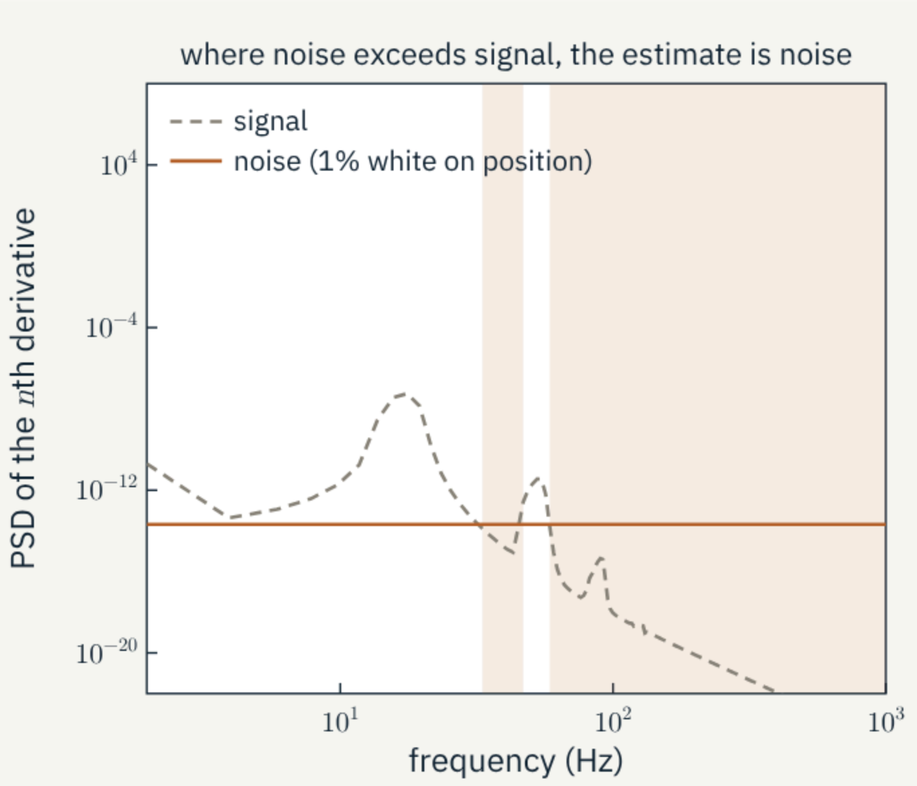
# Why it is ill-posed




Left: the amplification factor is unbounded at both ends. Right: the same effect on a real record — a 1% noise floor on position dominates the acceleration estimate above about 30 Hz.

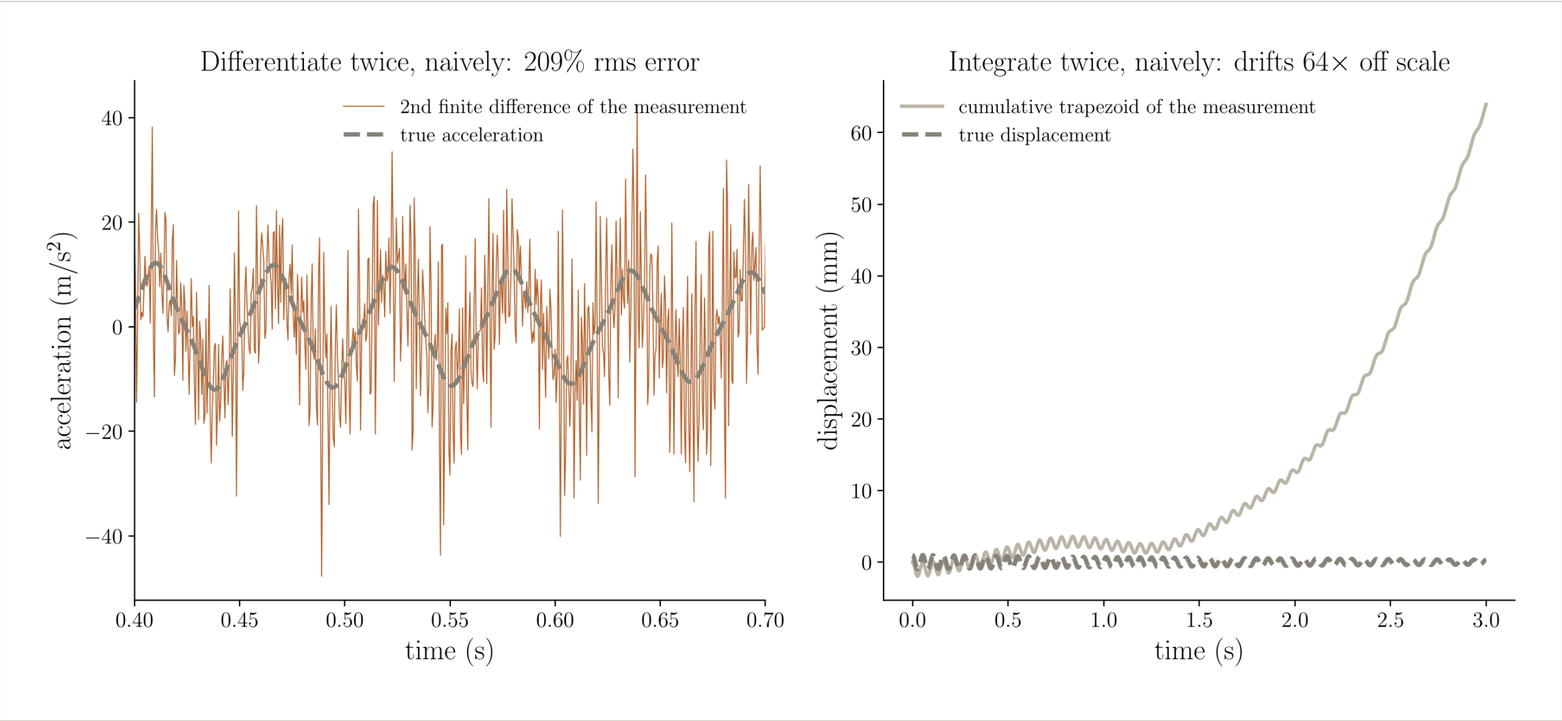
# Differentiation and integration: one gain, opposite ends



THE PROBLEM

# What happens if you ignore it




Left: second central difference of a 1% noisy position — 209% rms error. Right: plain cumulative trapezoid of an accelerometer with a 0.5% offset — 64x off scale in 3 s.

THE PROBLEM

# The only cure: state a prior

*Methods differ only in which regularity prior they impose — pick the row you actually believe*

| What you are willing to assume | The method it gives you |
|---|---|
| **The signal is bandlimited and periodic** | Spectral / FFT differentiation |
| **It is locally a low-order polynomial** | Savitzky–Golay, Lanczos |
| **It lies in a low-rank delay-embedding subspace** | Delay SVD (SSA) / SOD — now in the toolbox |
| **Its $p$-th derivative is small in $L_2$** | Whittaker–Henderson, Tikhonov, smoothing spline |
| **Its derivative is piecewise constant ($L_1$)** | Total variation (Chartrand) |
| **It is an integrated Wiener process** | Kalman / RTS smoother $\equiv$ Gaussian process |
| **Its third derivative is bounded by a known constant** | Levant sliding-mode (causal, online) |

# 01

# Differentiation methods



# Spectral (FFT) differentiation: strengths

*Multiply the spectrum by $(\mathrm{i}\omega)^n$, roll off above a cutoff, transform back.*

## STRENGTHS

- Exact for a bandlimited periodic signal — no discretisation error at all
- Most accurate of any method in the interior of a long record (0.54% velocity error here)
- Trivial to implement, $\mathcal{O}(N \log N)$, and the cutoff has an obvious physical meaning
- The natural choice when you already work in the frequency domain

# Spectral (FFT) differentiation: limitations

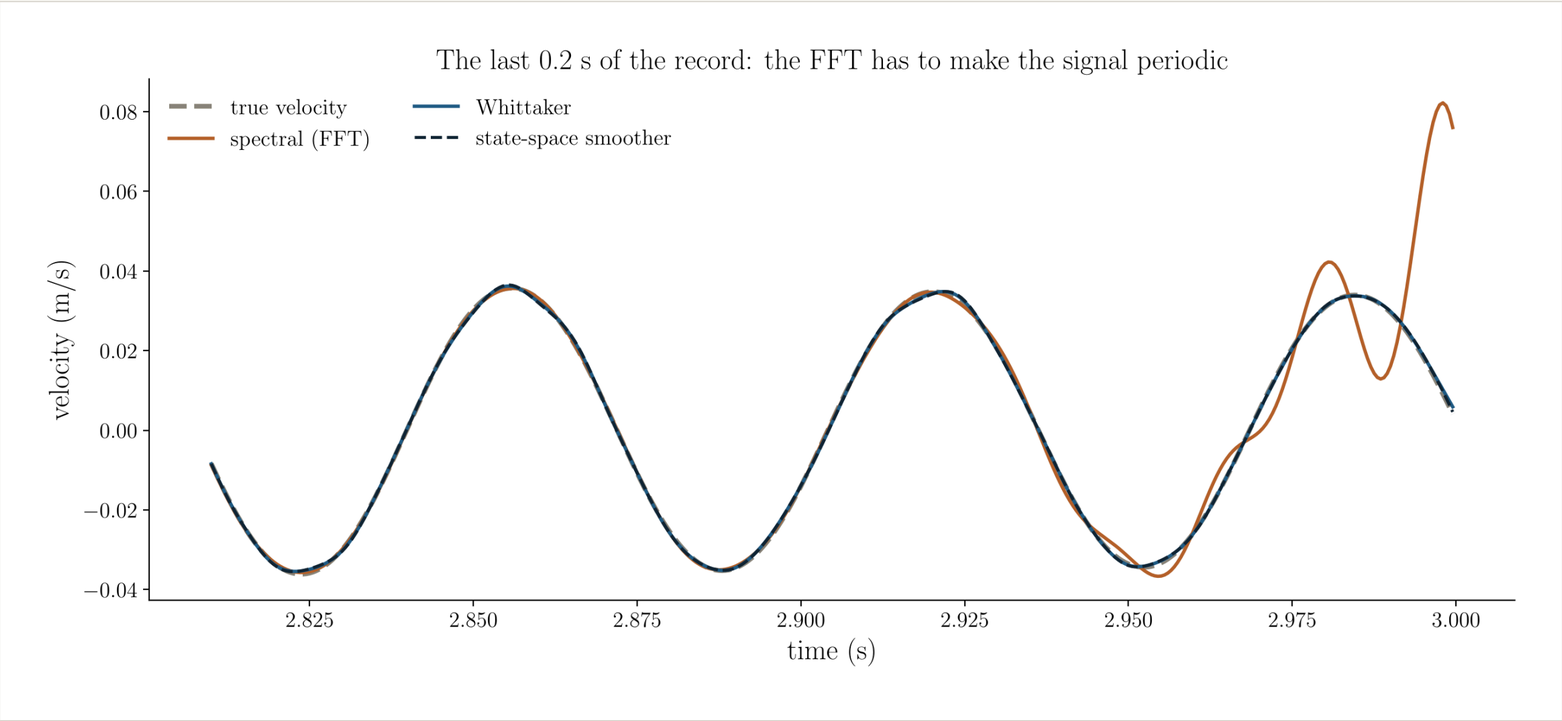
### LIMITATIONS

- Enforces periodicity — catastrophic at the record ends (next slide)
- Needs uniform sampling and a gap-free record
- Cutoff must be chosen; too high amplifies noise, too low removes signal
- Leakage if the record is not an integer number of periods (taper, then divide it out)

> **Use it when** The record is long, uniformly sampled, and you only care about the middle of it.

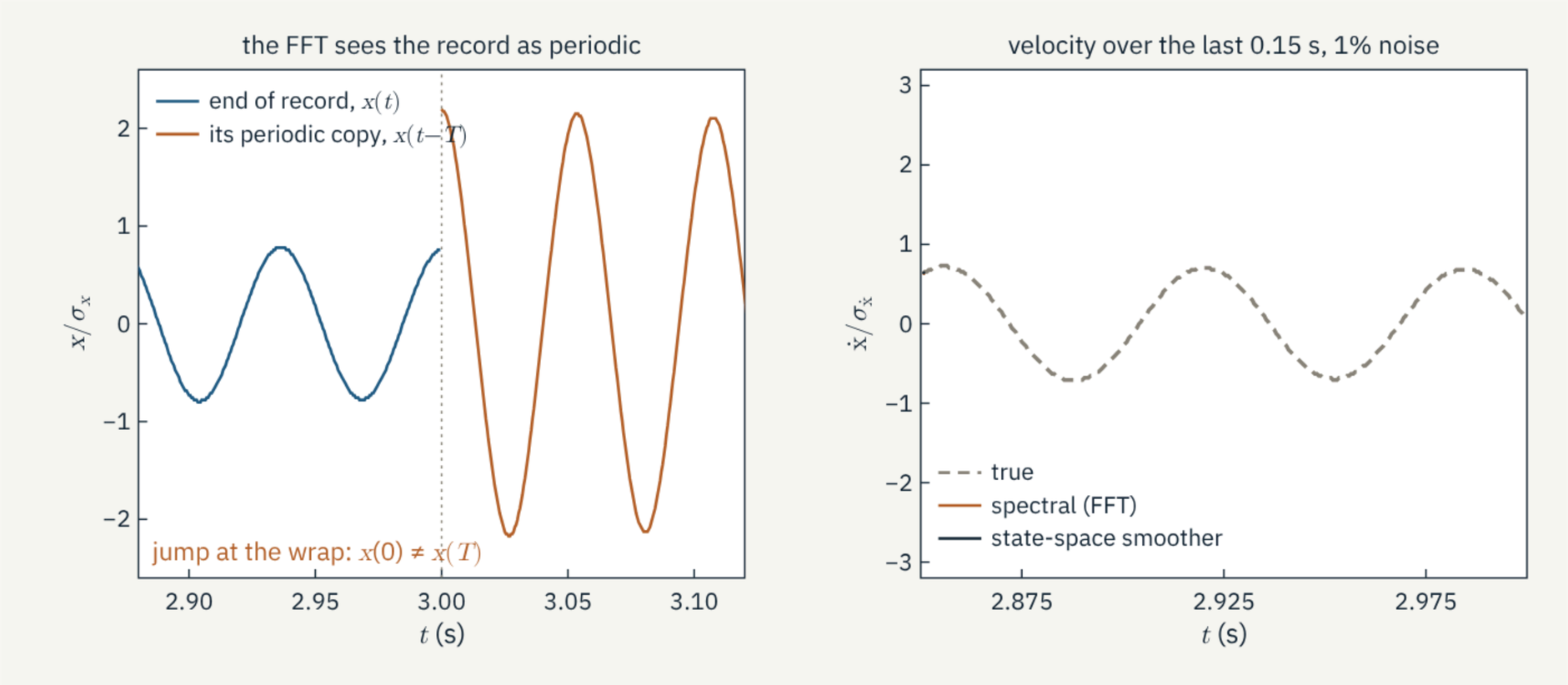
DIFFERENTIATION

# The FFT caveat that costs people real results




Velocity over the last 0.2 s of a 3 s free decay. Interior error 0.54%; over the full record, 8.4% — a 15x degradation.

# Why the FFT fails at the ends of the record






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# Savitzky–Golay: strengths

*Fit a low-order polynomial in a sliding window; differentiate the polynomial analytically.*

## STRENGTHS

- Simple, fast, and available in every toolbox
- No periodicity assumption — much better behaved at the ends than the FFT
- Preserves peak heights and widths better than a plain moving average
- Window length maps to a cutoff via Schafer’s rule, so it can be set from physics

# Savitzky–Golay: limitations

## LIMITATIONS

- Two knobs (window, polynomial order) that interact
- Implicitly assumes the signal is polynomial over the window — poor for lightly damped resonance
- Noticeably worse than an $L_2$ penalty for second derivatives (2.4% vs 0.9% here)
- Edge behaviour depends on the padding convention, which differs between implementations

**Use it when** You want something defensible in ten lines with no linear solve, and first derivatives are enough.

# Whittaker–Henderson and Tikhonov: strengths

*Penalise the $p$-th derivative in $L_2$:* $\min_s \|s - x\|^2 + \lambda \|D^p s\|^2$. *One banded solve.*

## STRENGTHS

- One sparse banded solve — $\mathcal{O}(N)$, no iteration, no convergence to check
- Flat accuracy across the whole record, ends included
- $\lambda$ has a closed-form map to a cutoff frequency, so no blind parameter sweep
- Same machinery runs the integration direction (Tikhonov) — one implementation, both tasks

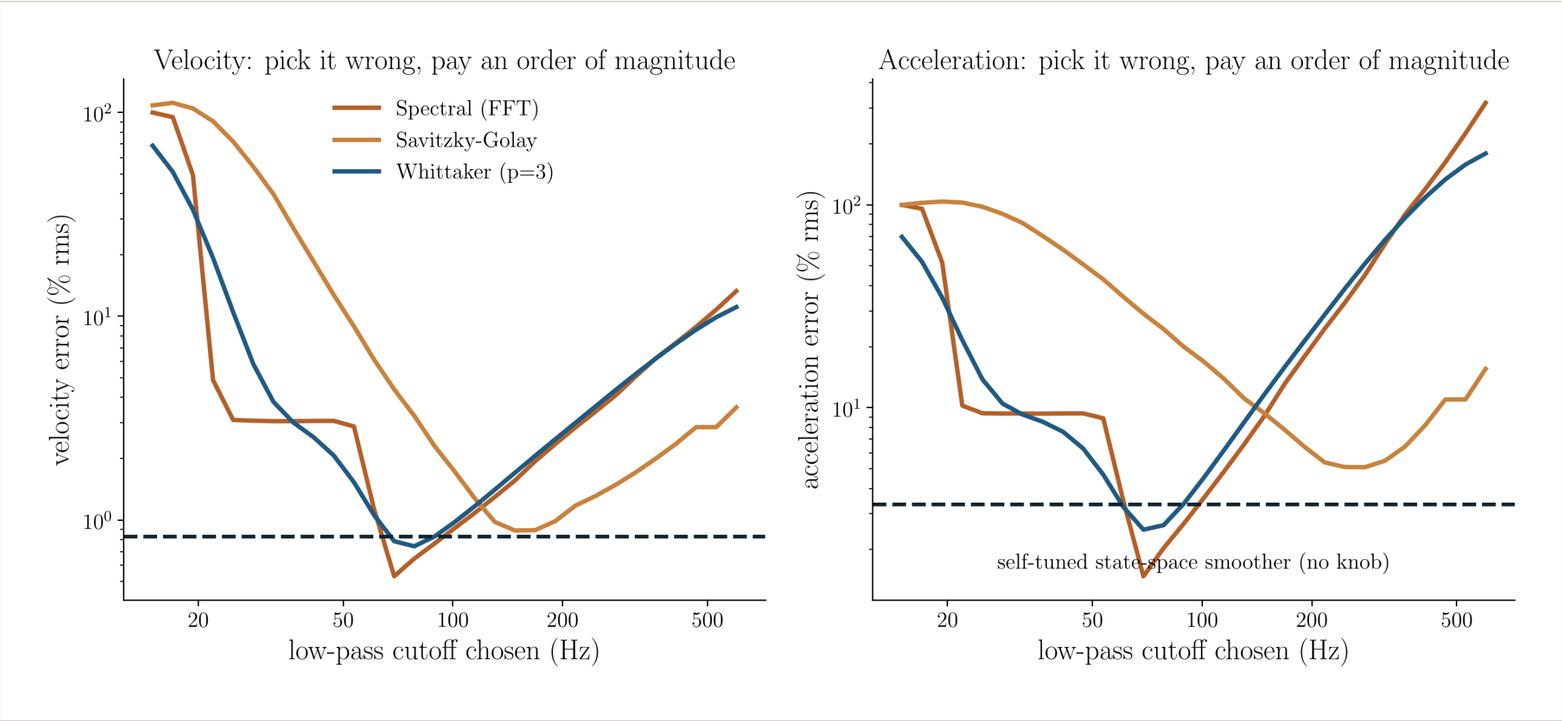
# Whittaker–Henderson and Tikhonov: limitations

### LIMITATIONS

- Still one knob, and the optimum is sharp for second derivatives
- Quadratic penalty rounds off genuine corners — wrong prior for impacting systems
- The normal matrix is badly scaled; use a sparse direct solver, not an iterative one
- No uncertainty estimate comes out of it

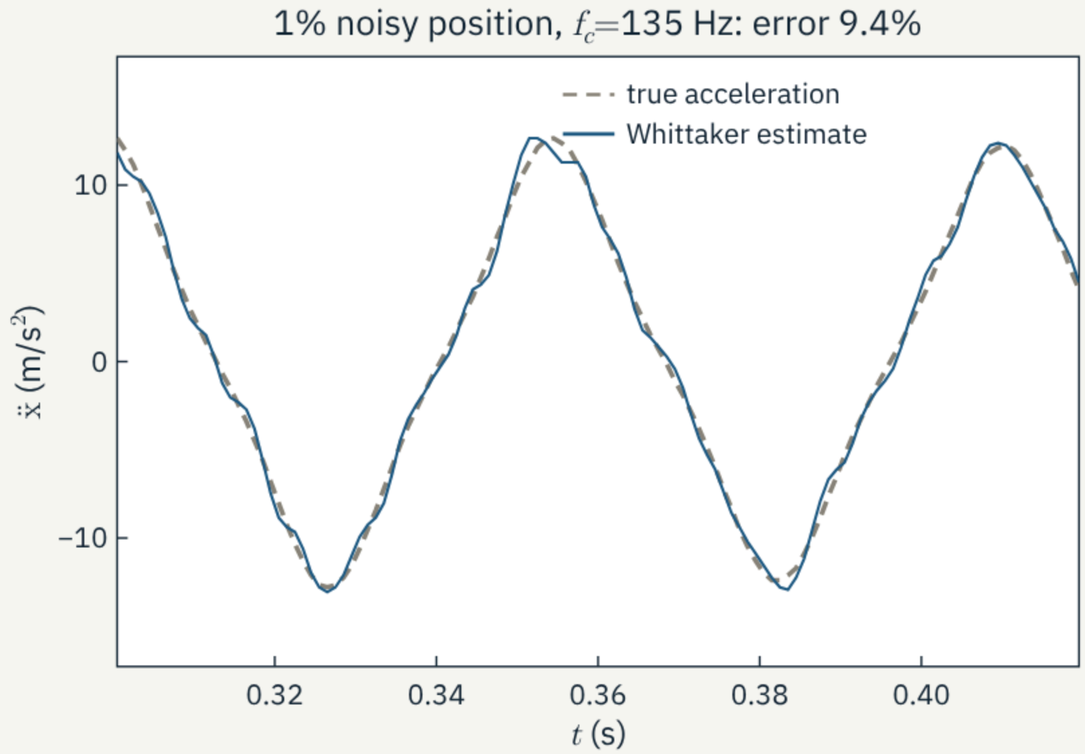
> **Use it when** This is the sensible default for smooth vibration data. Start here.

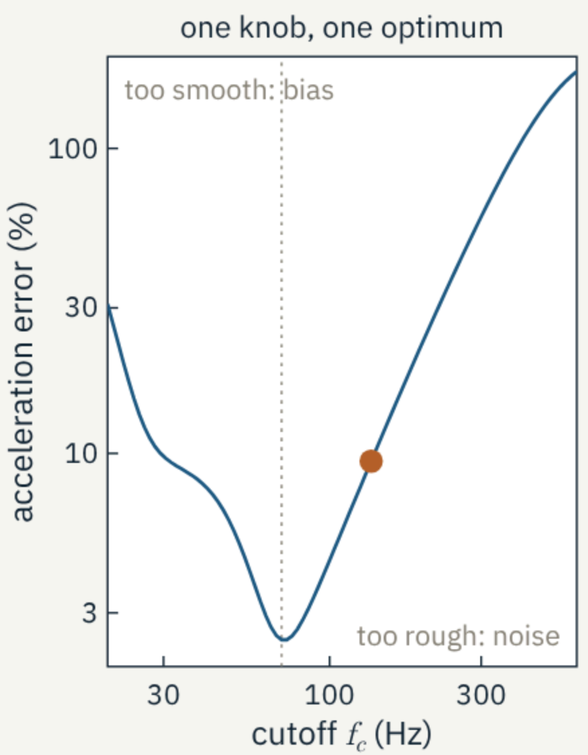
# Every method has an optimum you have to find




Error against exact ground truth as the cutoff is swept, 1% position noise. The dashed line is the self-tuned state-space smoother, which has no knob at all.

# One knob: sweeping the cutoff






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# Setting the knob from physics, not from a sweep

*Every $L_2$ method has a closed-form map from a corner frequency in Hz to its penalty weight*

| Method | Parameter | Rule | Set it from |
|---|---|---|---|
| **Whittaker–Henderson (order $p$)** | $\lambda$ | $\lambda = (2\pi f_c)^{-2p}$ | Highest mode you care about |
| **Tikhonov integration (order $n$)** | $\lambda$ | $\lambda = (2\pi f_c)^{2n}$ | Lowest frequency you trust |
| **Savitzky–Golay** | window $2M+1$ | $f_c\,\Delta t \approx \dfrac{k+1}{3.2M - 4.6}$ | Schafer's $-3$ dB rule |
| **Spectral** | cutoff | direct | Where the PSD meets the noise floor |
| **Total variation** | $\alpha$ | no closed form | Sweep; scale-sensitive, normalise first |
| **State-space smoother** | $q$ | fitted automatically | Marginal likelihood — nothing to set |

# Total variation: strengths

*Penalise $\|Du\|_1$ instead of $\|Du\|_2$. The $L_1$ norm does not punish a single large jump.*

## STRENGTHS

- The only family that preserves a genuine discontinuity in the derivative
- Best method tested at an impact: 16.9% vs 21–25% for every quadratic penalty
- Physically the right prior for impacts, stick–slip, backlash, dry friction reversal
- Recovers a piecewise-constant derivative exactly in the noise-free limit

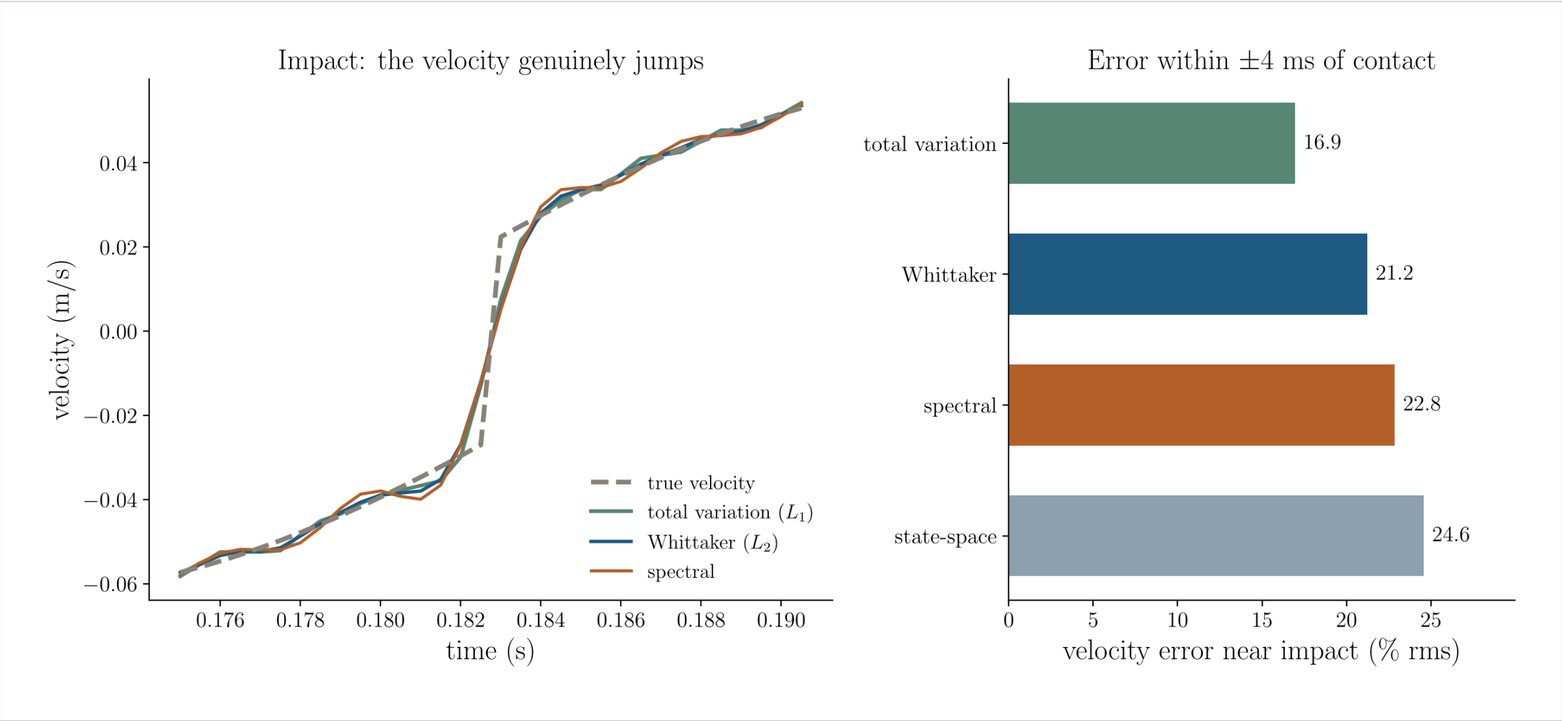
# Total variation: limitations

## LIMITATIONS

- Several times worse than an $L_2$ penalty on smooth signals — 5.1% vs 0.5% here
- Iterative, no closed form, and the objective is non-smooth
- $\alpha$ is tied to absolute signal scale — normalize before, rescale after
- The naive solver silently fails to converge (next slide)

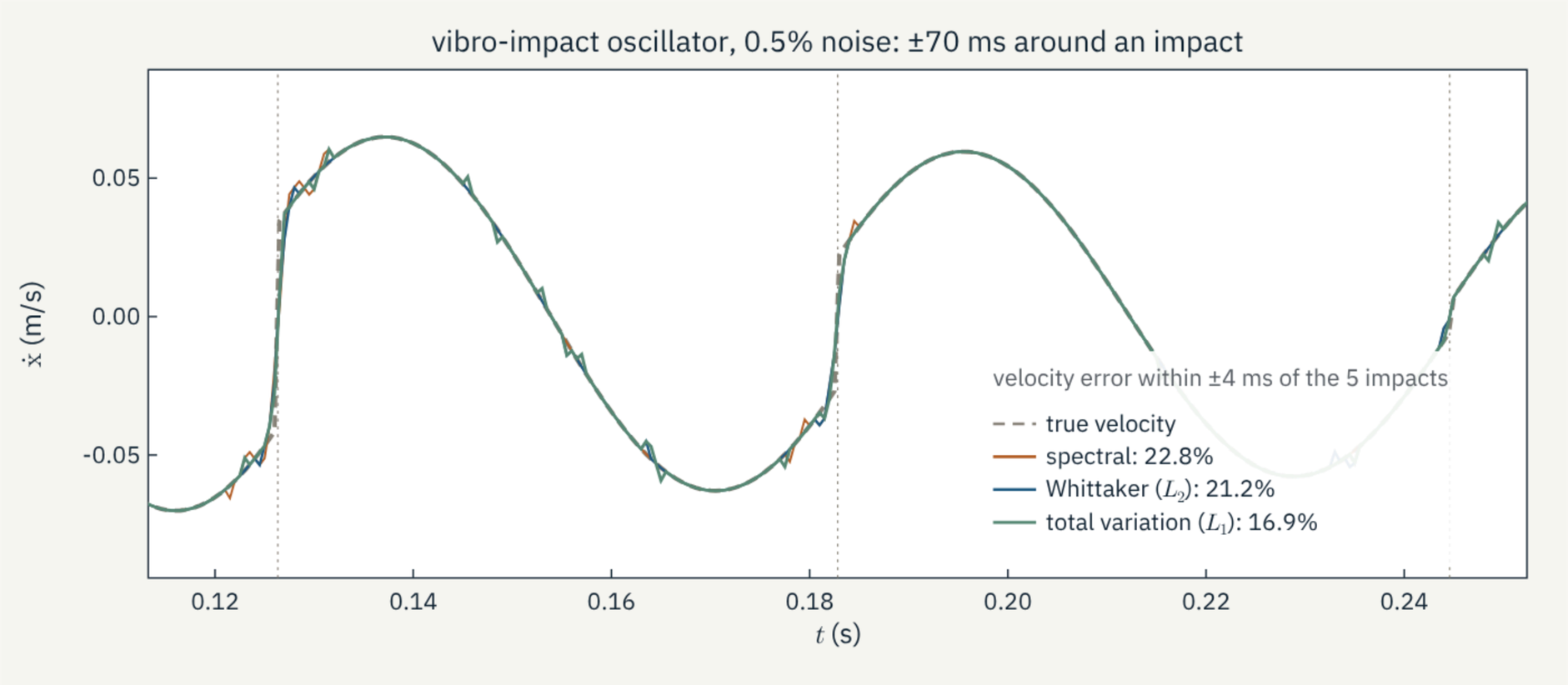
**Use it when** Your derivative genuinely jumps. Otherwise do not — it is the wrong prior for smooth data.

# Where total variation earns its place




Vibro-impact oscillator, restitution 0.85. Left: one impact, velocity jumps. Right: velocity error within ±4 ms of contact, best setting per method.

# At an impact, total variation tracks the jump most closely



# A cautionary tale: the solver was the result

*Reported numbers for an iterative method are claims about the solver as much as the estimator*

- Chartrand's formulation has a dense, severely ill-conditioned Hessian $A^{\mathsf{T}}A + \alpha L$.
- Conjugate gradients stalls at any iteration cap. IRLS on the smoothed $L_1$ stalls too — measured rate 0.997 per iteration.
- The unconverged iterate acts as extra smoothing and silently changes the answer.
- MATLAB's pcg warns. scipy's cg returns the same unconverged vector with no warning at all.
- Three successive solvers moved the same benchmark:

# Under-convergence flattered the method in both directions

| smooth signal | near impacts |
| --- | --- |
| 3.77% → 4.93% → **5.07%** | 17.9% → 16.6% → **19.2%** |

- Under-convergence had been flattering the method in both directions. Only the last pair is the true minimizer.
- Fix: substitute $w = Au$ to get $L_1$ trend filtering — the ADMM system is pentadiagonal and constant, so it factors once.

# Smoothing spline: scipy’s GCV choice returned a flat fit

*Found while porting to MATLAB; both are fixed in both languages*

## SMOOTHING SPLINE

- scipy’s make_smoothing_spline picks $\lambda$ by GCV. On the 6000-point decay it returned a flat fit: 100% error.
- The exact GCV score, with an $\mathcal{O}(N)$ trace, has a clear minimum: 1.18% velocity error. The failure was the implementation, not GCV.

# Levant: the default bound was 100× too small

**LEVANT DIFFERENTIATOR**

- Levant's default bound $L$ used the $(n+1)$-th derivative. The three-state form needs the third derivative bound for both outputs.
- $L$ was about 100× too small: 119% error. With max|jerk| from the state-space smoother: 7.7%, at the swept optimum.

Neither raised an error. Each returned a plausible-looking array.

Same lesson as the solver slide: check a default against a case where you know the answer.

# State-space: the Kalman / RTS kinematic smoother: strengths

*Model the signal as an integrated random walk; state = $(x,\ v,\ a,$ jerk, $\dots)$. Smooth, do not filter.*

## STRENGTHS

- Self-tuning — the single parameter $q$ is fitted by marginal likelihood, no sweep
- Handles gaps, non-uniform sampling and multirate channels natively (NaN = missing)
- Returns a posterior standard deviation — error bars, not just an estimate
- Does differentiation, integration AND sensor fusion with one implementation
- $\mathcal{O}(N)$, and equivalent to a Gaussian process without the $\mathcal{O}(N^3)$

# State-space: the Kalman / RTS kinematic smoother: limitations

## LIMITATIONS

- Costs something against a perfectly tuned method on very clean data (up to 3.7×)
- The prior is smoothness — it rounds corners just as the $L_2$ penalties do
- More code than a banded solve; the likelihood optimisation needs a sane bracket
- Bias states are weakly identified over short records — a drift absorber, not a calibration

**Use it when** You want one tool that covers every case, or you have two sensors, or you need uncertainty.

DIFFERENTIATION

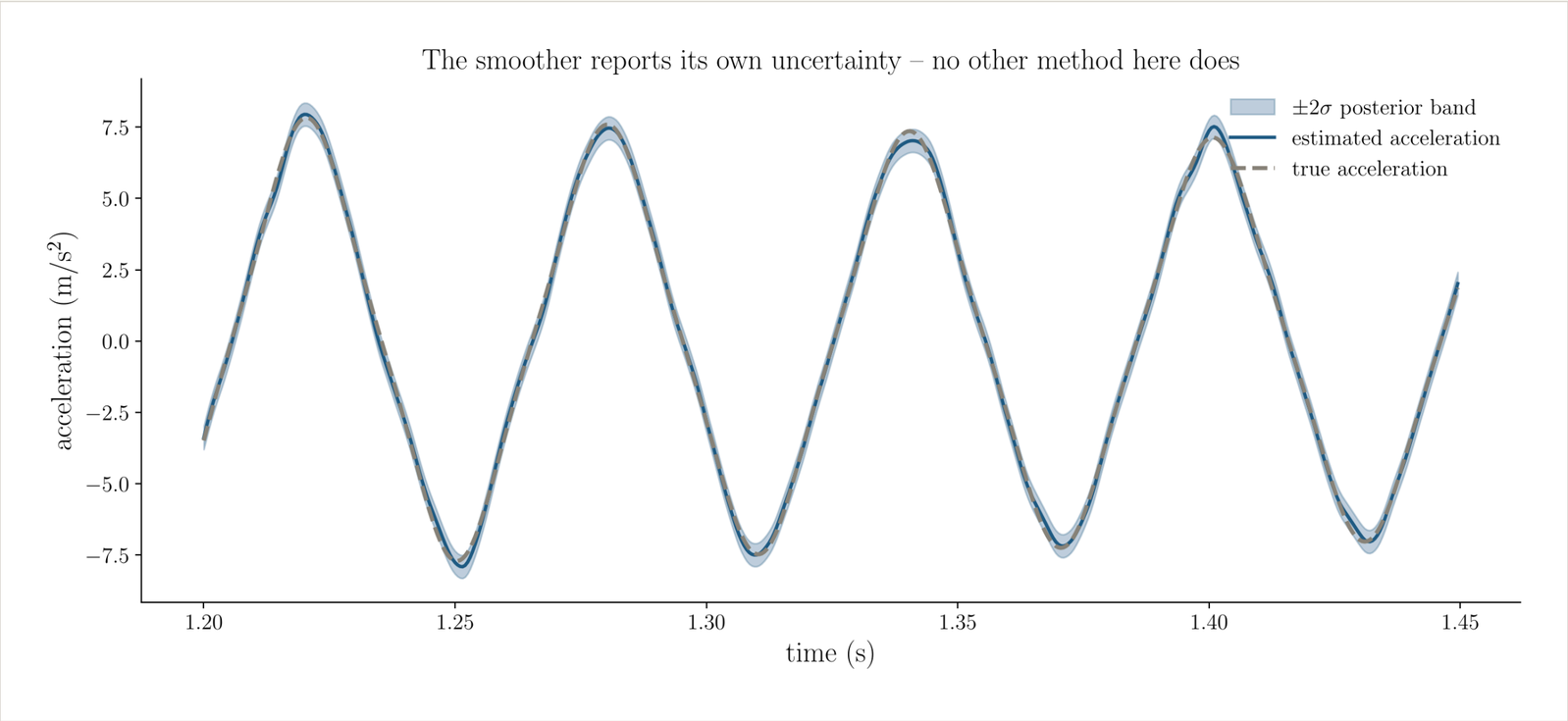
# What does self-tuning actually cost?




Ratio of the self-tuned error to the best error achieved by ANY method in this study with its knob tuned against ground truth.

DIFFERENTIATION

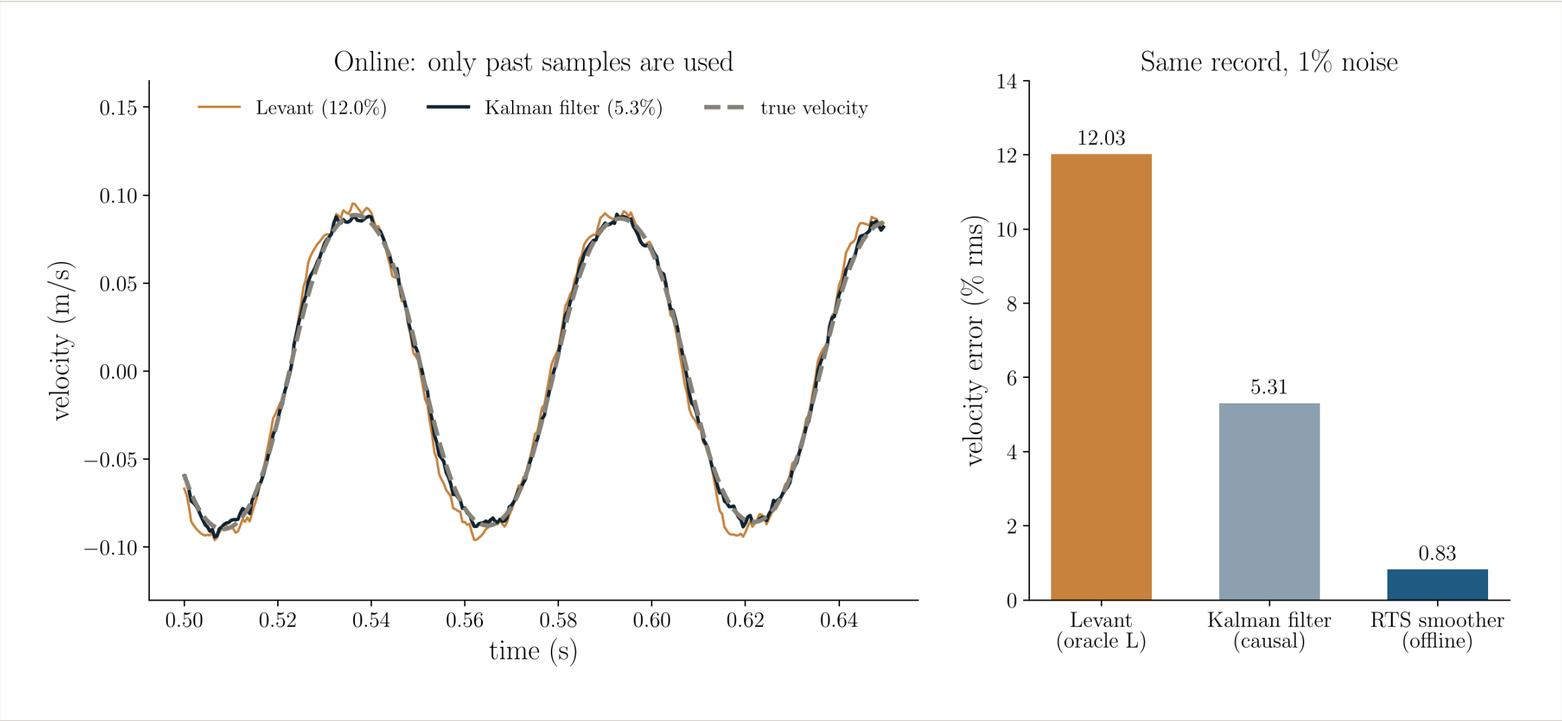
# It tells you how much to trust it




Acceleration from a noisy position record with its $\pm 2\sigma$ posterior band. No other method in this comparison produces one.

DIFFERENTIATION

# Online, the Kalman filter halves Levant’s error




Forward pass of the same state-space model, self-tuned: 5.3% velocity error against 12.0% for Levant at its best bound. Offline, the RTS smoother reaches 0.83%.

# 02

# Delay reconstruction



# Differentiate in the delay space, not on the raw record

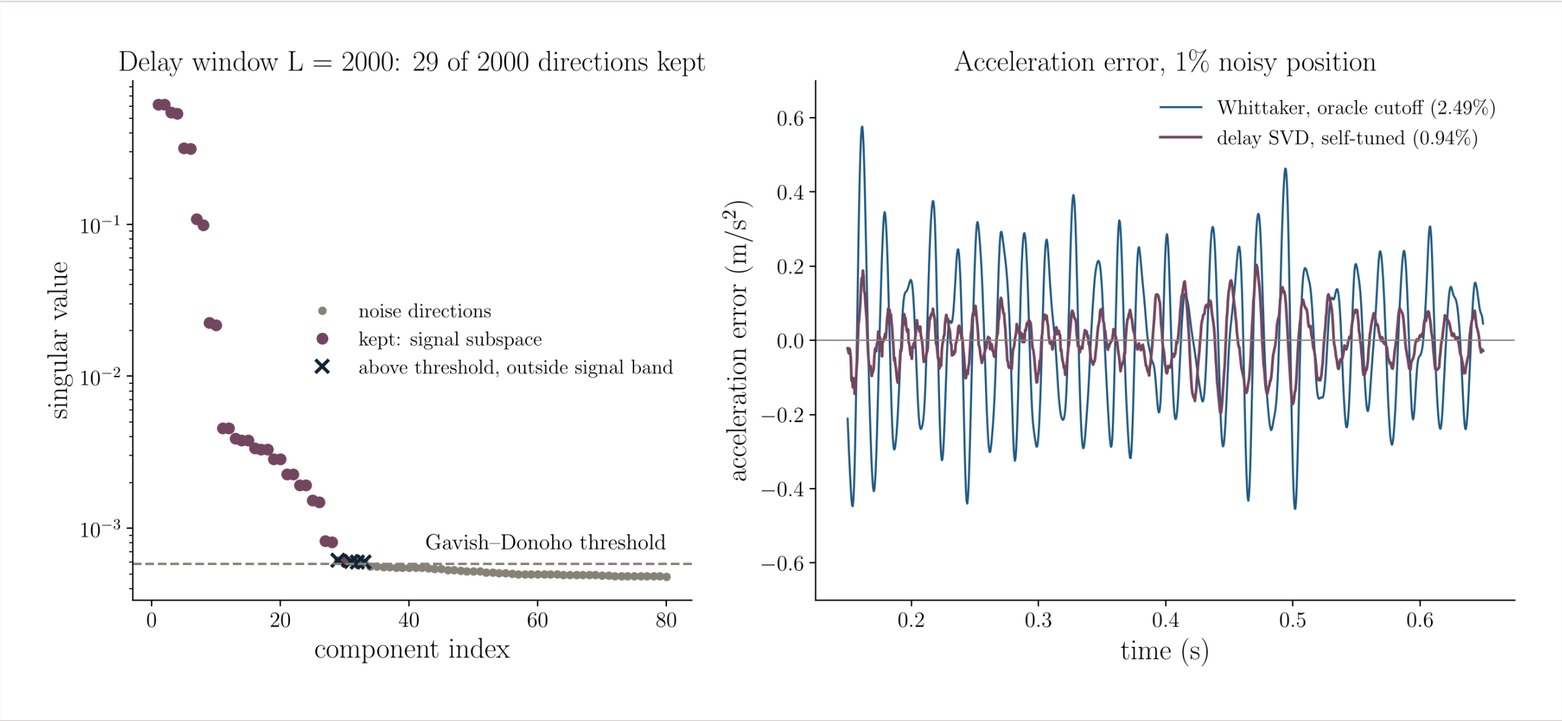
*Prior: the signal lies in a low-dimensional subspace of its delay reconstruction*

- **Embed:** rows of a Hankel matrix are delay vectors $(x_k, x_{k+1}, \dots, x_{k+L-1})$, window $L = N/3$.
- **Decompose:** SVD of that matrix; keep directions above the Gavish–Donoho hard threshold.
- **Guard:** drop any kept direction whose mean frequency lies above the band where the PSD clears the noise floor.
- **Reconstruct:** project, then average along anti-diagonals. Several sensors embed side by side.
- **Differentiate** the reconstruction with a fourth-order stencil. It is band-limited, so the stencil error is negligible.
- No knob: $L$ from the record length, dimension from the threshold, guard from the spectrum.



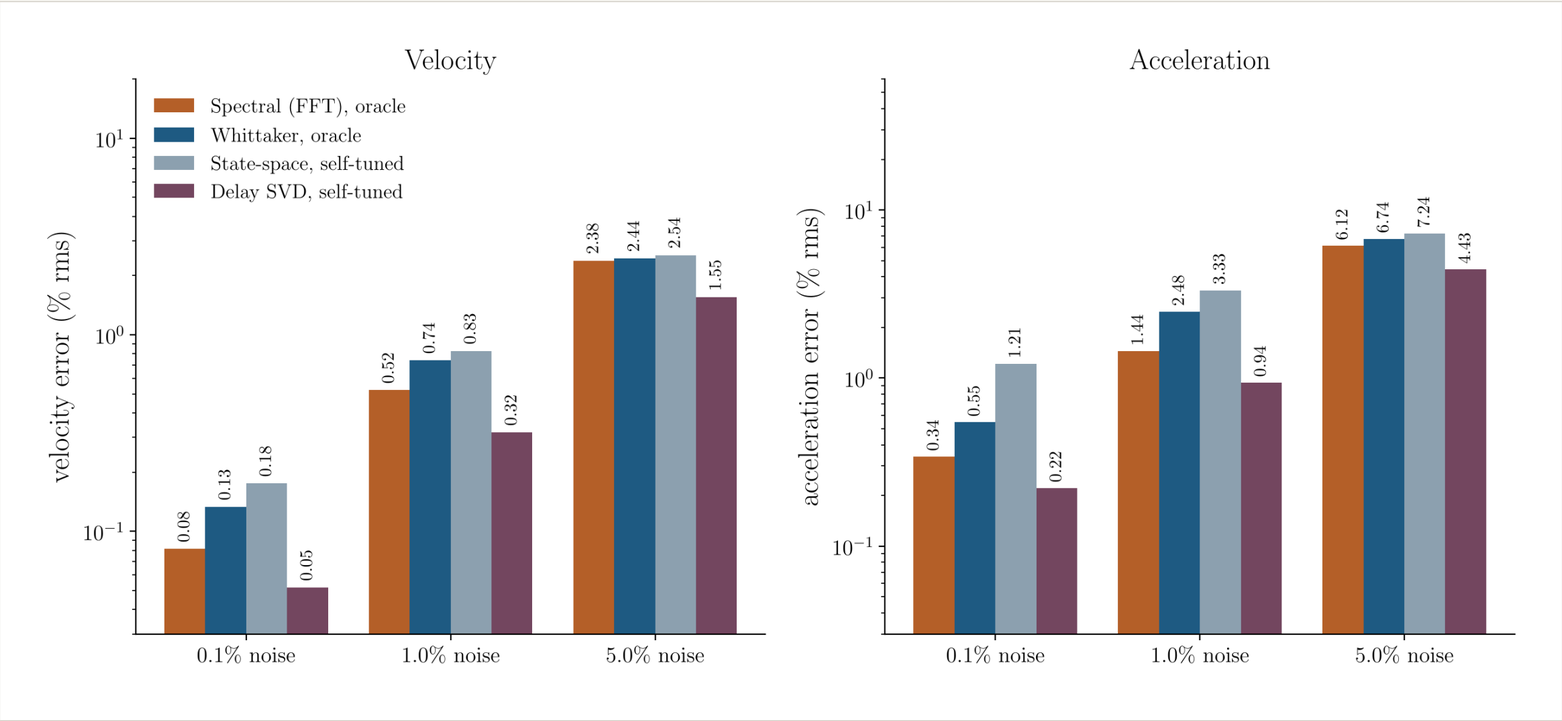
DELAY RECONSTRUCTION

# 29 of 2000 delay directions carry the decay




Left: singular spectrum of the trajectory matrix, 1% position noise. Right: acceleration error of the self-tuned delay estimate against Whittaker at its oracle cutoff.

# Self-tuned delay SVD beats every oracle-tuned method




Duffing free decay, same noise realisations as the main benchmark. Oracle = cutoff swept against the exact answer. The delay method sees no truth.

# Delay reconstruction (SSA): strengths

*Project onto the signal subspace of the trajectory matrix; differentiate or integrate the reconstruction.*

## STRENGTHS

- No knob: window from record length, dimension from the hard threshold
- Best accuracy of any method tested on decays, by 1.4–1.6× at every noise level
- Several sensors share one subspace (multichannel SSA)
- Full-record accuracy: no FFT-style edge damage

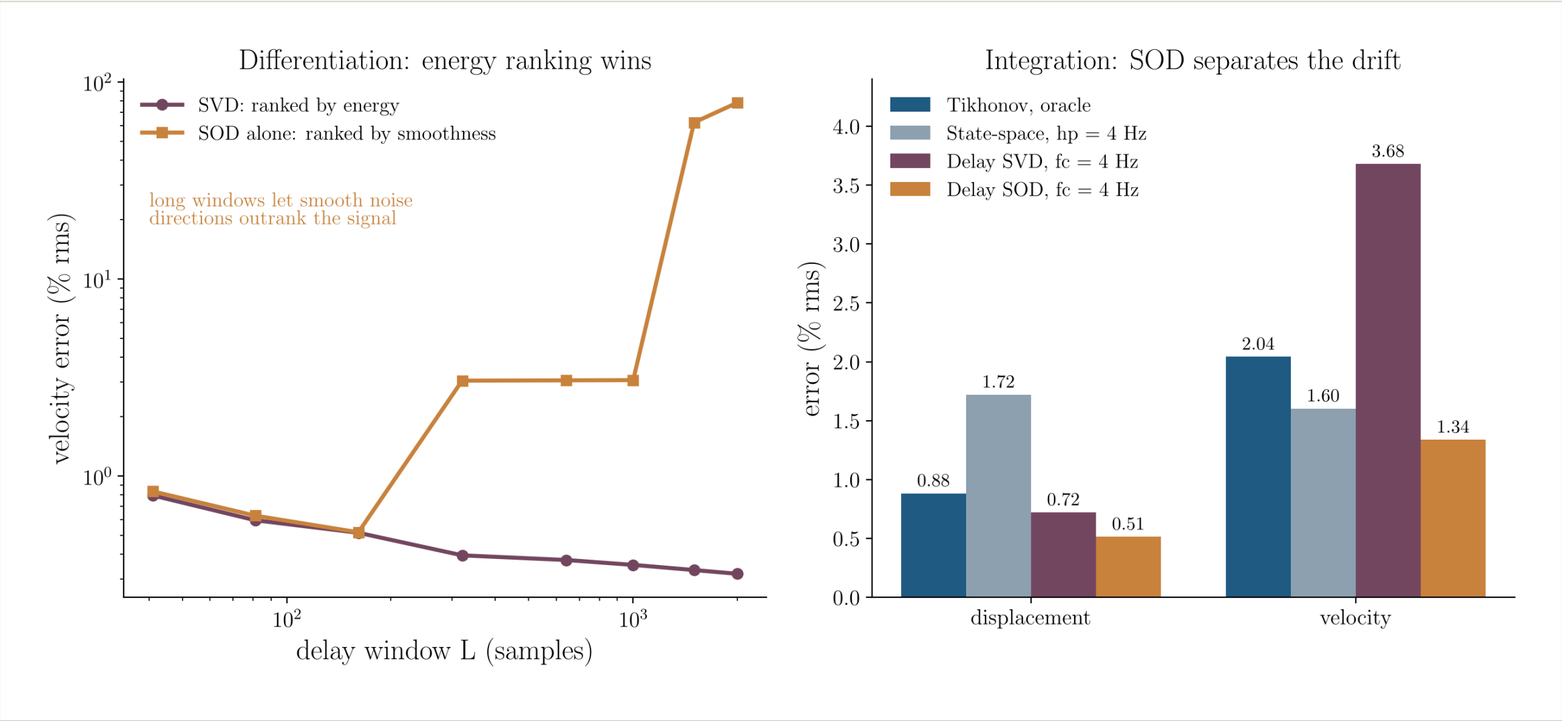
# Delay reconstruction (SSA): limitations

## LIMITATIONS

- The prior is quasi-periodicity: no gain at impacts (23% vs 17% for TV)
- Cost grows as $L^2 N$: about 1 s for $N = 6000$, $L = 2000$
- Needs uniform sampling and a gap-free record
- No error bars, and integration still needs a corner

**Use it when** The record is a free decay, a ringdown or a steady forced response.

# SVD for differentiation, SOD for integration




Left: SOD alone ranks directions by smoothness, so long windows admit smooth noise. Right: SOD inside the SVD subspace orders it by frequency and isolates the drift.

# Where delay reconstruction helps, and where it does not

*Velocity / acceleration error, % rms; integration rows give displacement / velocity*

| Record | Best competitor | Delay method | Verdict |
|---|---|---|---|
| Duffing decay, 1% noise | 0.52 / 1.44 spectral, oracle | 0.32 / 0.94 SVD | helps |
| 3-DOF chain, mass 3 | 0.95 / 2.21 Whittaker, oracle | 0.38 / 0.64 SVD | helps |
| 3-DOF chain, 3 sensors jointly | 0.95 / 2.21 Whittaker, oracle | 0.26 / 0.63 MSSA | small extra gain |
| Vibro-impact, near impacts | 17.0 total variation | 23.1 SVD | does not help |
| Drifting accelerometer | 0.88 / 2.04 Tikhonov, oracle | 0.51 / 1.34 SOD | helps |
| Drifting accelerometer | 0.50 / 0.25 spectral, oracle | 0.51 / 1.34 SOD | spectral wins here |

# 03

# Integration and fusion

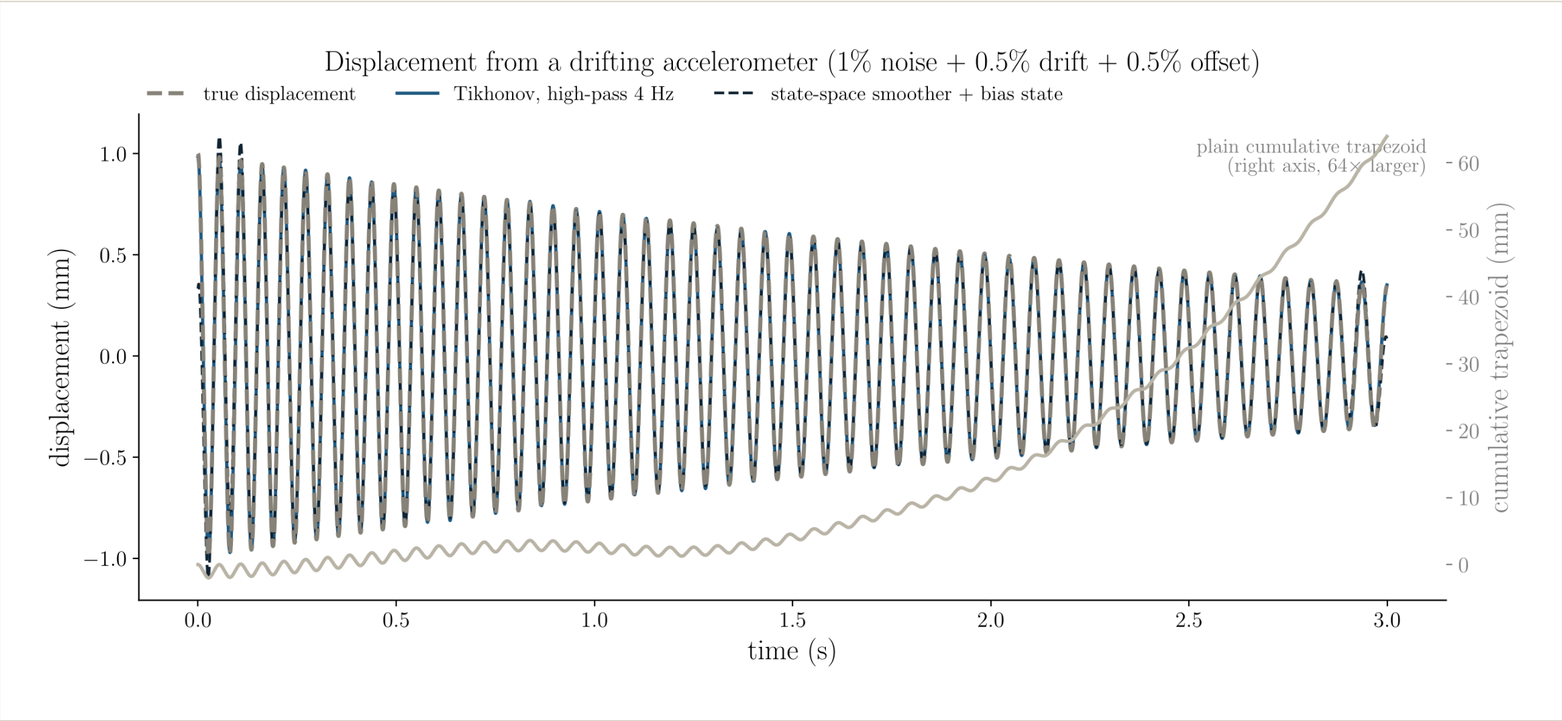


# Integration has an extra problem: the constants

*Two integration constants are unobservable — every method makes an assumption, so make yours explicit*

- Acceleration determines displacement only up to $x_0 + v_0 t$. Those two constants are not in the data.
- A DC offset of 0.5% integrates twice into a parabola that swamps the signal within seconds.
- Every working method therefore imposes a high-pass — the question is whether it admits to it.
- **Detrend + high-pass filter:** fast, but the corner is chosen by eye and distorts the low modes.
- **Tikhonov:** the corner appears explicitly as $\lambda = (2\pi f_c)^{2n}$. One banded solve.
- **State-space with a bias state:** the corner becomes a prior on the unobservable directions.
- Choose the corner below your lowest mode of interest and above the drift band — and say what you chose.
- If you also have a displacement channel, you do not need the corner at all. That is the next slide.

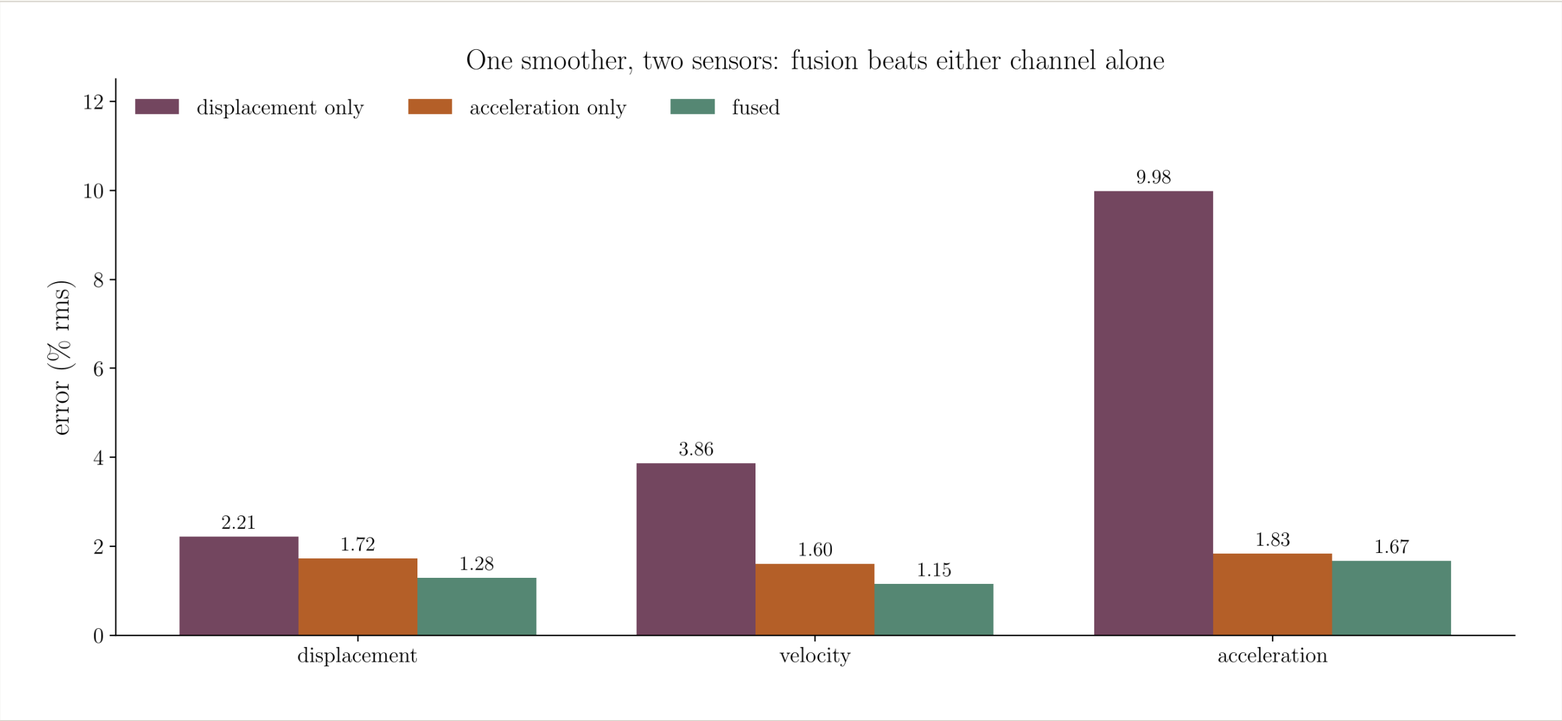
# Integration done three ways




Displacement recovered from an accelerometer with 1% noise, 0.5% drift and a 0.5% offset. Regularised methods overlay the truth; the plain trapezoid needs its own axis.

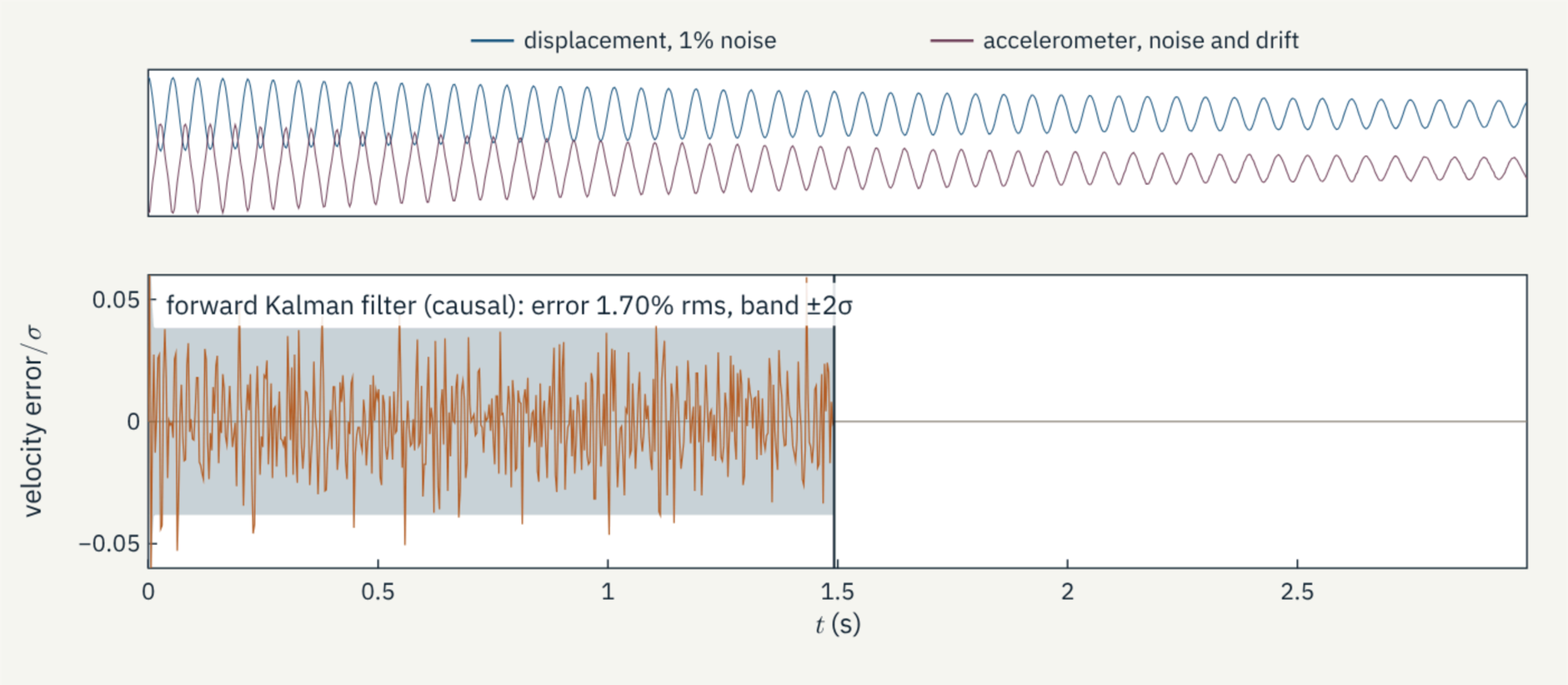
INTEGRATION

# If you have both sensors, use both at once




One state-space smoother, 200 Hz displacement at 3% noise fused with 2 kHz acceleration at 1% noise plus drift.

# Forward filter, then backward smoother



# 04

# In practice



IN PRACTICE

# The comparison: five classical methods

| Method | Prior | Knobs* | Ends | Gaps | Error bars | Best at |
|---|---|---|---|---|---|---|
| **Spectral (FFT)** | bandlimited | 1 | poor | no | no | long uniform records |
| **Savitzky–Golay** | local polynomial | 2 | fair | no | no | quick first derivatives |
| **Whittaker / Tikhonov** | small $p$-th deriv. | 1 | good | no | no | smooth data, both directions |
| **Smoothing spline** | small $p$-th deriv. | 1 | good | yes | no | non-uniform sampling |
| **Total variation** | piecewise-const. | 1 | good | no | no | impacts, stick–slip |

**Knobs = free parameters you must choose; three methods now set their own*

IN PRACTICE

# The comparison: smoother, delay and online methods

| Method | Prior | Knobs* | Ends | Gaps | Error bars | Best at |
|---|---|---|---|---|---|---|
| **State-space smoother** | integrated Wiener | 0 | good | yes | yes | everything else, and fusion |
| **Delay SVD / SOD (new)** | low-rank delay space | 0 | good | no | no | free decays, ringdowns |
| **Kalman filter, causal (new)** | integrated Wiener | 0 | n/a | yes | yes | online / control loops |
| **Levant sliding mode** | bounded 3rd deriv. | 1 | n/a | no | no | finite-time exactness |

**Knobs = free parameters you must choose; three methods now set their own*

IN PRACTICE

# What to actually do

*A decision list, in the order the questions actually arise*

- Smooth data, one sensor, want it simple → **Whittaker–Henderson**, cutoff from physics.
- Smooth data, want no knob or need uncertainty → **state-space smoother**, nstate = 4 (5 if very clean).
- Quasi-periodic record (free decay, ringdown, steady forced response) → **delay SVD**, no knob.
- Two sensors → **always fuse them in one smoother**. Never process them separately and reconcile after.
- Impacts, backlash, stick–slip → **total variation**, and check that the solver converged.
- Integrating acceleration alone → **Tikhonov or a bias-state smoother**; report the high-pass corner.
- Online or in a control loop → **the causal Kalman filter**; Levant only when you need finite-time exactness with a known bound.
- Never: a bare finite difference, or a bare cumulative trapezoid.

IN PRACTICE

# Validating on a record with no ground truth

*Four checks that cost minutes and one habit that costs nothing*

Everything shown here was measured against an exact solution. On your data you have none — so:

- **Cross-check.** Differentiate the estimated displacement twice; compare with the measured acceleration in the band you trust.
- **Read the posterior band.** If the feature you are interpreting is narrower than $\pm 2\sigma$, it is not there.
- **Move the knob.** Change the cutoff by a factor of two either way; if the answer moves more than the band, report that.
- **Hold out a channel.** With two sensors, fit on one and predict the other — the honest test.
- **State your choices.** Method, cutoff or high-pass corner, and solver convergence, alongside the sampling rate.

IN PRACTICE

# Summary

- Differentiating displacement and integrating acceleration are one ill-posed problem, not two.
- Noise amplification is a property of the operator. Every method you can choose adds a prior instead.
- So the real question is never ‘which filter’ — it is ‘what do I believe about this signal?’
- For smooth vibration data: a quadratic penalty, tuned from a cutoff in hertz.
- For a quasi-periodic record: project onto its delay-space subspace first. Self-tuned, it beat every oracle-tuned method here.
- For a genuine corner: an $L_1$ penalty, with the solver’s convergence checked.
- For anything else — gaps, two sensors, no time to tune, or a need for error bars: the state-space smoother.
- And if you have displacement and acceleration, fuse them. It is strictly better and it removes a knob.

# Implementation

Python and MATLAB implementations of every method here, with matching names.

```
kin_diff_whittaker, kin_diff_spectral, kin_diff_savgol, kin_diff_tvreg,
kin_diff_spline, kin_diff_gp, kin_diff_levant

kin_int_tikhonov, kin_int_spectral, kin_int_trapz_detrend

kin_delay_project, kin_diff_delay, kin_int_delay — delay reconstruction, SVD or SOD

kin_smooth — differentiation, integration, fusion; causal option for online use
```

Cross-validated between the two languages: 38 of 38 checks pass, agreement 1e-16 to 1e-4.

**Cite:** `kinematics.cite()` or `kin_cite` prints BibTeX · doi:10.5281/zenodo.23549236

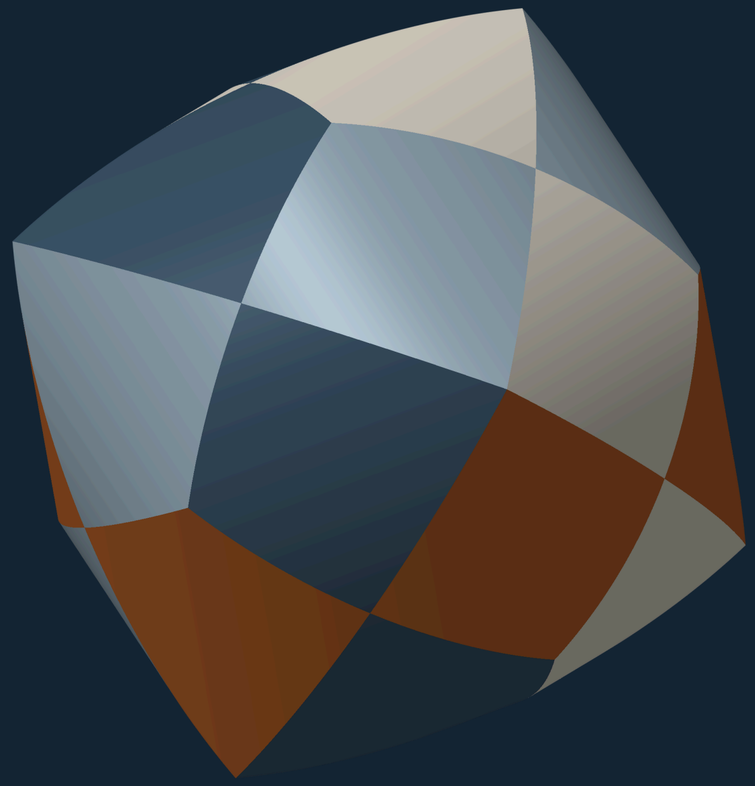
# References

*Primary sources for each family, not a full bibliography*


- Chartrand (2011), Numerical differentiation of noisy, nonsmooth data — *ISRN Appl. Math.*
- Eilers (2003), A perfect smoother — *Anal. Chem.* 75:3631
- Särkkä & Solin, *Applied Stochastic Differential Equations*, CUP 2019
- Kim, Koh, Boyd & Gorinevsky (2009), $\ell_1$ trend filtering — *SIAM Review* 51(2):339
- Levant (1998), Robust exact differentiation via sliding mode technique — *Automatica* 34:379
- Broomhead & King (1986), Extracting qualitative dynamics from experimental data — *Physica D* 20
- Golyandina, Nekrutkin & Zhigljavsky (2001), *Analysis of Time Series Structure: SSA* — CRC
- Gavish & Donoho (2014), The optimal hard threshold for singular values is $4/\sqrt{3}$ — *IEEE Trans. Inf. Theory*
- Chelidze & Zhou (2006), Smooth orthogonal decomposition-based vibration mode identification — *J. Sound Vib.*

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# Questions

When amplitude changes the story

github.com/nld-uri/kinematics-toolbox · doi:10.5281/zenodo.23549236

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